Dye Concentration Evolution (Advection-Diffusion)
CFL=0.05, Diffusion Number=0.125

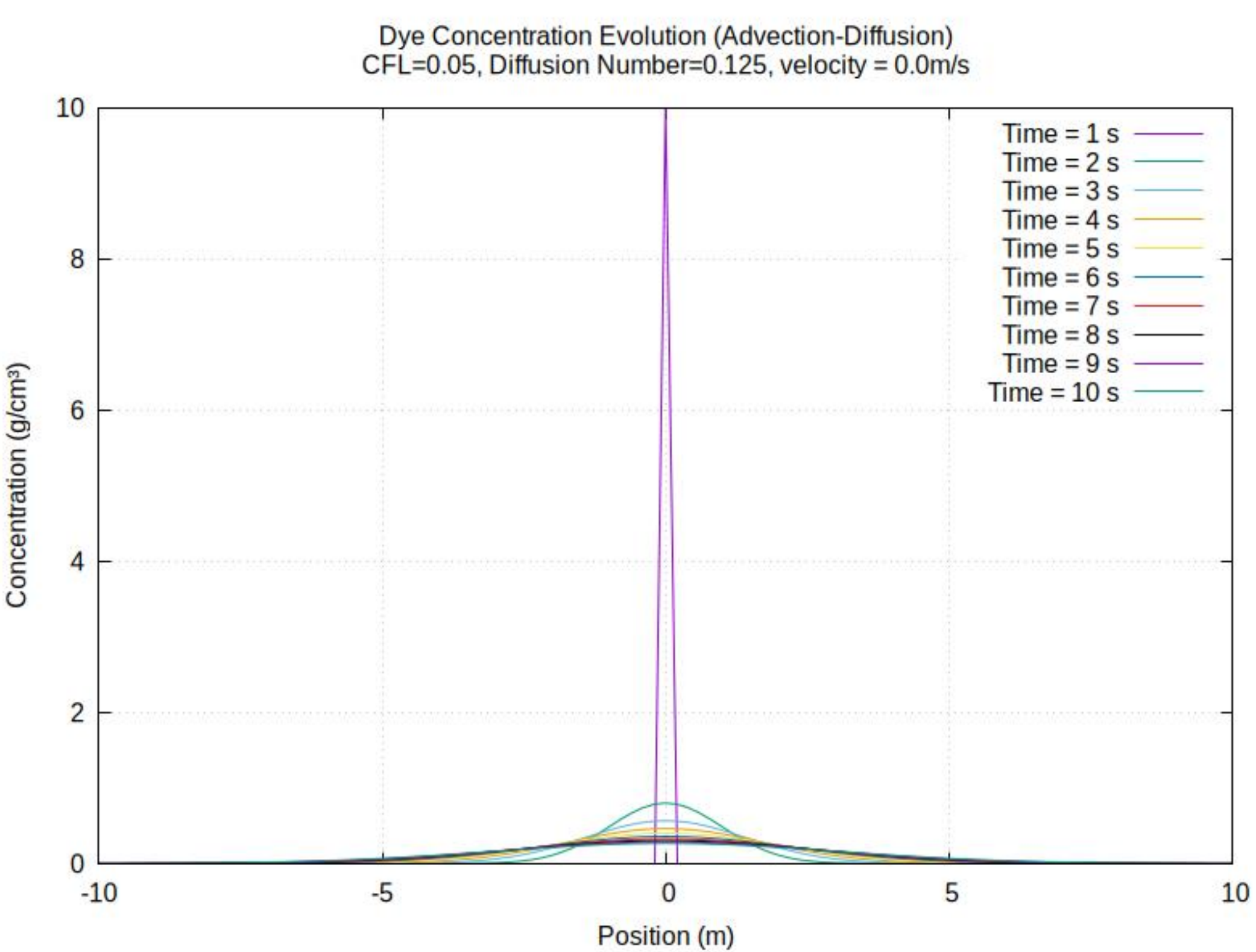
10 s

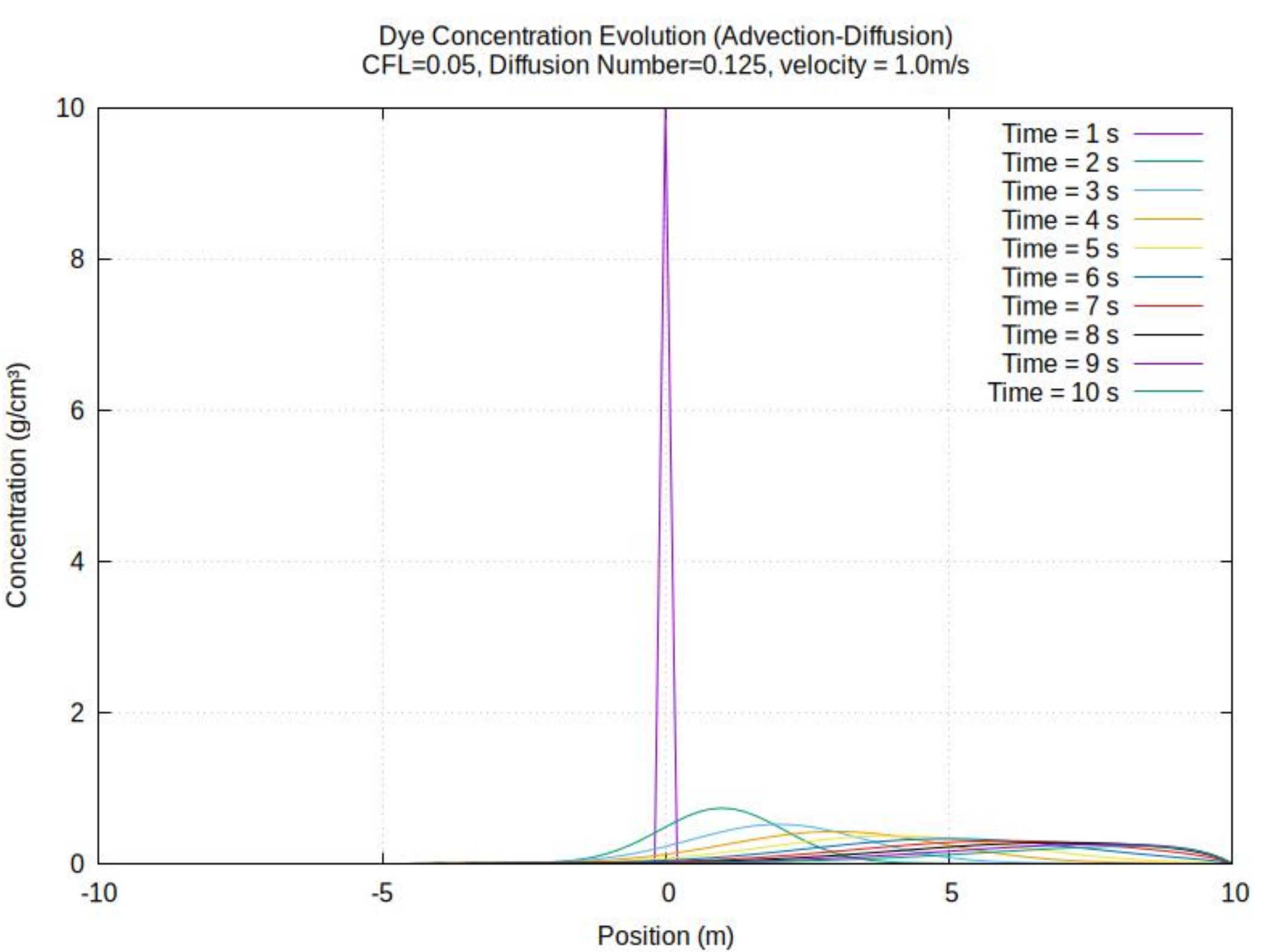
Concentration (kg/m³)

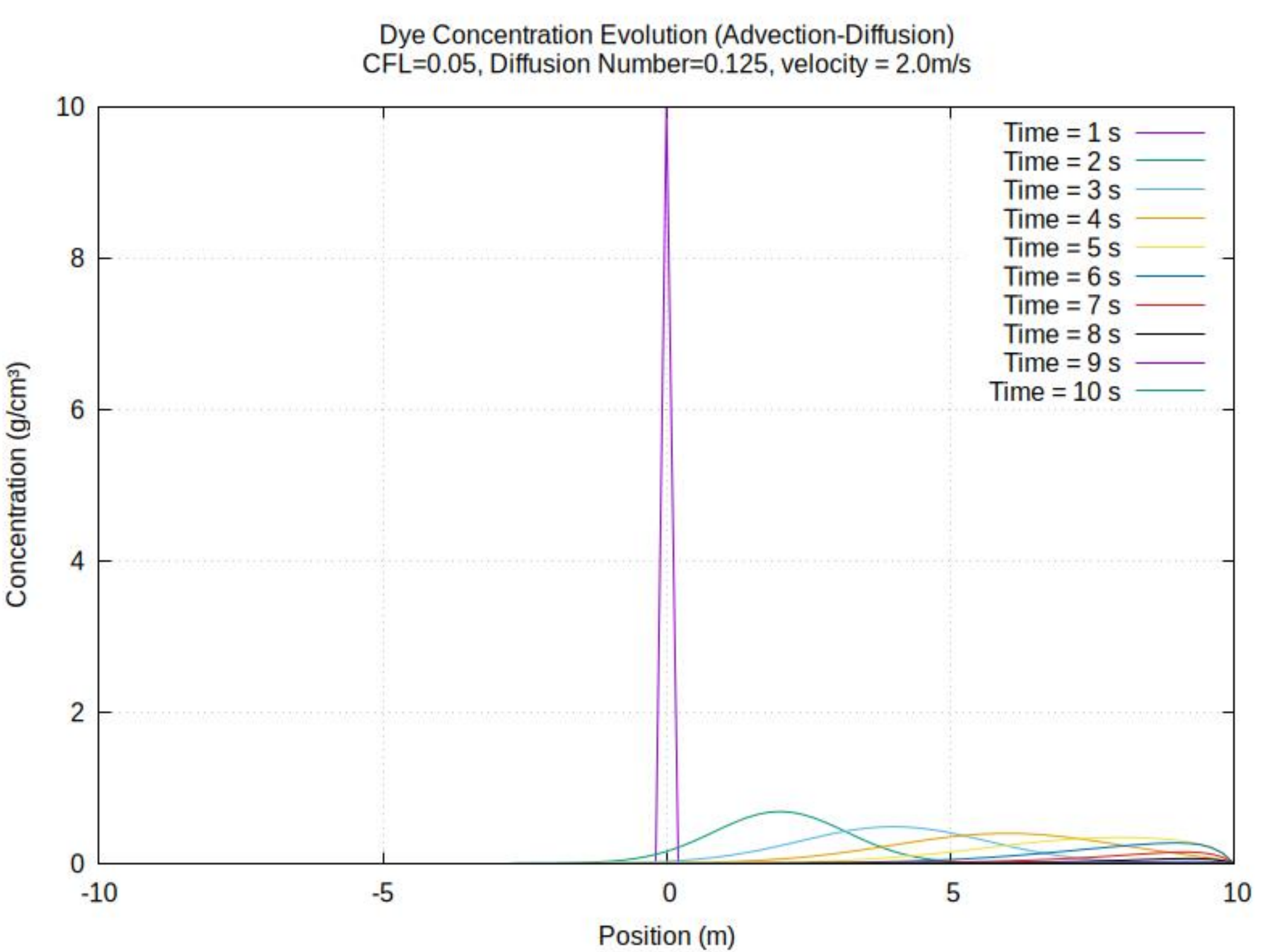
12000
10000
8000
6000
4000
2000
0

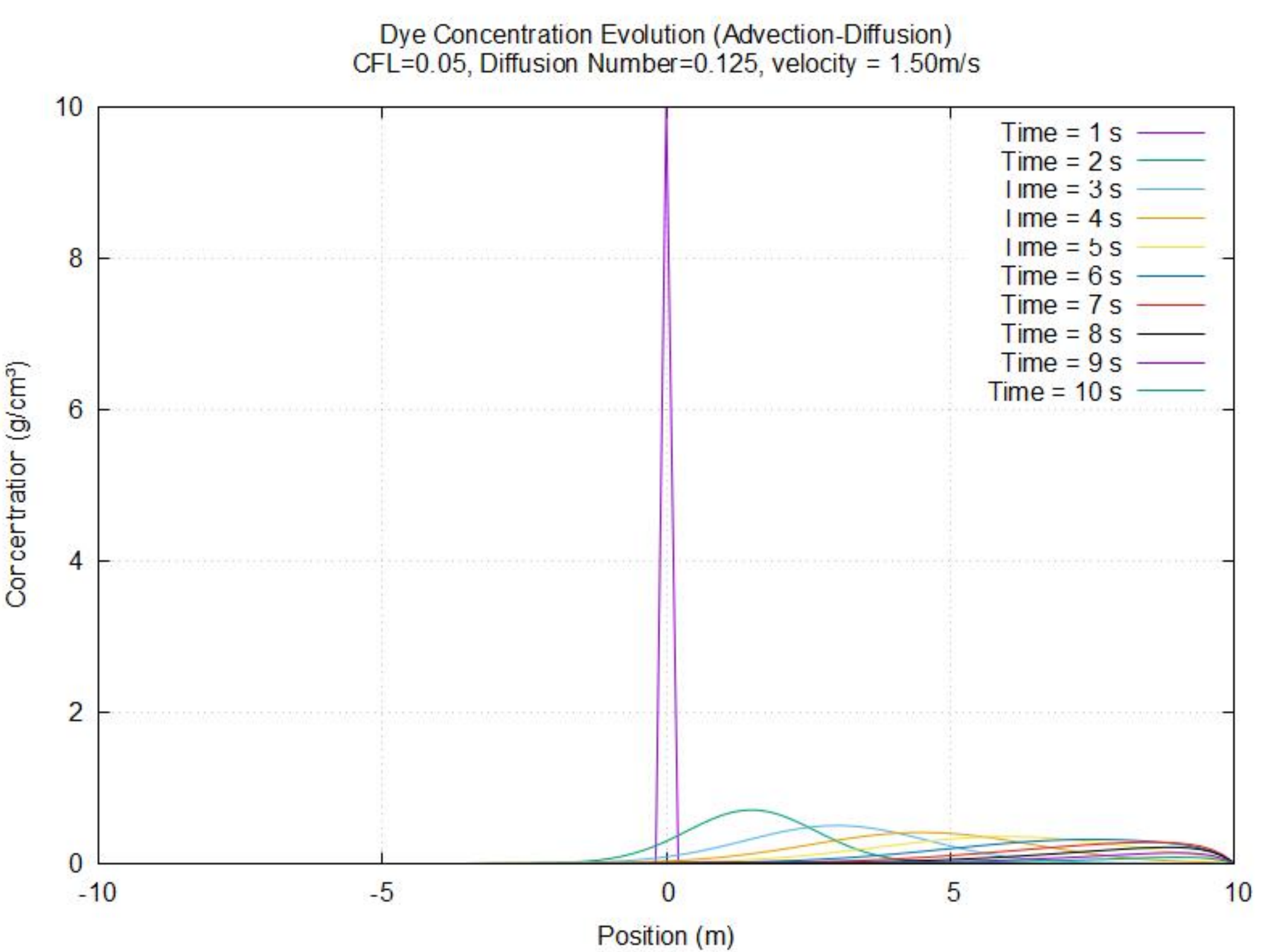
-10
-5
0
5
10

Position (m)


Dye Concentration Evolution (Advection-Diffusion)
CFL=0.05, Diffusion Number=0.125, velocity = 0.0m/s
Concentration (g/cm³)
Position (m)
10
8
6
4
2
0
-10
-5
0
5
10
Time = 1 s
Time = 2 s
Time = 3 s
Time = 4 s
Time = 5 s
Time = 6 s
Time = 7 s
Time = 8 s
Time = 9 s
Time = 10 s


Dye Concentration Evolution (Advection-Diffusion)
CFL=0.05, Diffusion Number=0.125, velocity = 1.0m/s
Concentration (g/cm³)
Position (m)
10
8
6
4
2
0
-10
-5
0
5
10
Time = 1 s
Time = 2 s
Time = 3 s
Time = 4 s
Time = 5 s
Time = 6 s
Time = 7 s
Time = 8 s
Time = 9 s
Time = 10 s


Dye Concentration Evolution (Advection-Diffusion)
CFL=0.05, Diffusion Number=0.125, velocity = 2.0m/s
Concentration (g/cm³)
Position (m)
10
8
6
4
2
0
-10
-5
0
5
10
Time = 1 s
Time = 2 s
Time = 3 s
Time = 4 s
Time = 5 s
Time = 6 s
Time = 7 s
Time = 8 s
Time = 9 s
Time = 10 s


Dye Concentration Evolution (Advection-Diffusion)
CFL=0.05, Diffusion Number=0.125, velocity = 1.50m/s
Time = 1 s
Time = 2 s
Time = 3 s
Time = 4 s
Time = 5 s
Time = 6 s
Time = 7 s
Time = 8 s
Time = 9 s
Time = 10 s
Concentration (g/cm³)
Position (m)
10
8
6
4
2
0
-10
-5
0
5
10